TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

# Determination of the Rigidity Modulus of the Material of a Wire by the Dynamical (Oscillation) Method

*A disc hung from the wire twists back and forth — the period of torsional oscillation measures how stiffly the material resists shear.*

PHY-1104 · PHYSICS-I SESSIONAL · EXPERIMENT 5 · LEVEL 1 TERM 1

## 1. Aim and theory

**Aim:** to find the rigidity (shear) modulus η of a wire's material from the period of a torsional pendulum. A heavy disc (or bar) of known moment of inertia I hangs from the wire (length L, radius r). Twisted and released, it oscillates with period T. The wire's restoring torque for twist θ is $\pi\eta r^4\theta/(2L)$, and equating the motion to simple harmonic motion gives

$$T = 2\pi \sqrt{( 2\, I\, L / (\pi\, \eta\, r^4) )} \;\Rightarrow\; \eta = 8\pi\, I\, L / (T^2\, r^4)$$

This is the **dynamical** method — the modulus is extracted from a timing measurement rather than from a static twist — and it is exquisitely sensitive to r: $\eta \propto 1/r^4$, so the radius measurement dominates everything.

## 2. Apparatus and procedure

- **Apparatus:** the experimental wire clamped vertically, a uniform disc (or a bar with two equal sliding masses) suspended from its lower end, stopwatch, screw gauge, metre rule, balance and calipers for the disc.
- Determine I of the suspended body: for a solid disc of mass M and radius R about its central axis, $I = \tfrac{1}{2}MR^2$. (For a bar with masses, I is computed from its geometry or calibrated with an added known ring.)
- Measure L (clamped length of wire) and r (screw gauge, several places, perpendicular pairs — then average; raise to the fourth power only at the end).
- Twist the disc through a small angle, release without pushing, and time 20 complete oscillations; repeat three times; compute T.
- Compute η. A valuable check: repeat with a second length L′ — $T^2$ should scale in proportion to L.

## 3. Worked calculation (model data)

**RESULT**

Suspended body (disc + chuck), about the wire's axis: $I = 0.0500$ kg m², built up from the parts' masses and dimensions as in §2. Wire: $L = 0.500$ m; radius $r = 0.50$ mm $= 5.0 \times 10^{-4}$ m $\rightarrow r^4 = 6.25 \times 10^{-14}$ m⁴. Period: 20 oscillations in 224 s $\rightarrow T = 11.2$ s; $T^2 = 125.4$ s².

$\eta = 8\pi IL/(T^2 r^4) = (8\pi \times 0.0500 \times 0.500)/(125.4 \times 6.25 \times 10^{-14}) = 0.6283/(7.84 \times 10^{-12}) =$ **$8.0 \times 10^{10}$ Pa** — the accepted rigidity modulus of steel.

*Note:* I must always be the moment of inertia of everything that oscillates — disc, chuck and any attached masses — about the wire's axis; omitting the chuck is a quiet systematic error in this experiment.

## 4. Precautions

- Measure the wire's radius with extreme care and at many points — $\eta \propto 1/r^4$ turns a 1% radius error into a 4% modulus error.
- Release the disc without lateral push: it must purely twist; swinging or wobbling contaminates the period.
- Keep the twist amplitude modest and constant between runs; count oscillations at a fixed reference mark.
- The wire must hang vertically and be clamped firmly at both ends — a slipping chuck adds false length and false period.
- Time many oscillations (20 or more) so stopwatch reaction error divides away.

## 5. Practice (viva preparation)

Answer in your own words; brief model answers follow.

1. Why is this called the dynamical method, and what is the static alternative it avoids?
2. With $I = 0.0400$ kg m², $L = 0.600$ m, $r = 0.45$ mm and $T = 12.6$ s, compute $\eta$.
3. The wire is replaced by one of twice the radius, same length and material, same disc. What happens to T?
4. Why does L appear to the first power but r to the fourth in the formula?
5. A student measures the wire's diameter as 1.00 mm but uses 1.00 mm as the radius. By what factor is $\eta$ wrong?
6. How would you check, within this experiment, that the period formula's L-dependence is behaving?

## Answer key

1. Because the modulus is obtained from the dynamics of oscillation — a period — rather than from a static twist under a known torque. The static method needs an accurate couple and an optical lever to read a tiny twist; the dynamical method trades those for timing, which is precise and simple.
2. $r^4 = (4.5 \times 10^{-4})^4 = 4.10 \times 10^{-14}$ m⁴; $T^2 = 158.8$; $\eta = (8\pi \times 0.0400 \times 0.600)/(158.8 \times 4.10 \times 10^{-14}) = 0.6032/(6.51 \times 10^{-12}) = 9.3 \times 10^{10}$ Pa — a little above steel's value; in the laboratory such a result points first at the radius measurement (Q5 shows why).
3. $T^2 \propto 1/r^4$ at fixed I and L, so doubling r cuts $T^2$ by 16 and T by 4: the new period is 11.2/4 = 2.8 s on the model apparatus. (This sensitivity is why torsional wires in instruments are so fine.)
4. Because the geometry of torsion makes the restoring torque proportional to $r^4/L$: each shell of the wire contributes shear resistance weighted by its radius, and integrating over the cross-section yields the fourth power of r, while a longer wire simply twists more per unit torque, in direct proportion to L.
5. Using d in place of r makes $r^4$ too big by $2^4 = 16$, so η comes out 16 times too small — the single most destructive slip in this experiment, and the reason the formula is written with r while the screw gauge measures d.
6. Repeat the timing at a second clamped length L′: $T^2$ should scale in exact proportion ($T'^2/T^2 = L'/L$) with the same disc and wire. A failure of proportionality exposes a slipping clamp or a wrong I before the final value is believed.

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